

# Solutions and Rubrics for Q1 and Q2

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## Rubrics

### Q1 Rubric (5 marks)

- 5.00 marks – Both truth tables fully correct (all columns, no mistakes).
- 4.00 marks – Minor mistake in one truth table, rest correct.
- 3.00 marks – Minor mistakes in both truth tables but structure correct.
- 2.00 marks – Only partial tables or major mistakes, but some valid work shown.
- 1.00 marks – Attempt made but largely incorrect.
- 0.00 marks – No valid attempt.

### Q2 Rubric (5 marks)

- 5.00 marks – Proof is complete, logically consistent, and clearly shows why equivalent propositions have the same DNF (using either truth table or algebra, or both).
- 4.00 marks – Mostly correct proof but missing some explanation or formal justification.
- 3.00 marks – Partial reasoning, captures main idea but lacks clarity/detail.
- 2.00 marks – Attempted explanation but logically weak or incomplete.
- 1.00 marks – Very little relevant work.
- 0.00 marks – No valid attempt.

## Solutions

### Q1(a) Solution

We need to compute the truth table for  $(p \Rightarrow q) \wedge \neg q \Leftrightarrow r$ .

Truth Table:

| p | q | r | $p \Rightarrow q$ | $\neg q$ | $(p \Rightarrow q) \wedge \neg q$ | $(p \Rightarrow q) \wedge \neg q \Leftrightarrow r$ |
|---|---|---|-------------------|----------|-----------------------------------|-----------------------------------------------------|
| T | T | T | T                 | F        | F                                 | F                                                   |
| T | T | F | T                 | F        | F                                 | T                                                   |
| T | F | T | F                 | T        | F                                 | F                                                   |
| T | F | F | F                 | T        | F                                 | T                                                   |
| F | T | T | T                 | F        | F                                 | F                                                   |
| F | T | F | T                 | F        | F                                 | T                                                   |

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| F | F | T | T | T | T | T |
| F | F | F | T | T | T | F |

### Q1(b) Solution

We need to compute the truth table for  $(p \oplus \neg q) \vee \neg r \Rightarrow p$ .

Truth Table:

| p | q | r | $\neg q$ | $p \oplus \neg q$ | $(p \oplus \neg q) \vee \neg r$ | $(p \oplus \neg q) \vee \neg r \Rightarrow p$ |
|---|---|---|----------|-------------------|---------------------------------|-----------------------------------------------|
| T | T | T | F        | T                 | T                               | T                                             |
| T | T | F | F        | T                 | T                               | T                                             |
| T | F | T | T        | F                 | F                               | T                                             |
| T | F | F | T        | F                 | T                               | T                                             |
| F | T | T | F        | F                 | F                               | T                                             |
| F | T | F | F        | F                 | F                               | T                                             |
| F | F | T | T        | T                 | T                               | F                                             |
| F | F | F | T        | T                 | T                               | F                                             |

### Q2 Solution

We are asked to prove: If two propositions are equivalent, then they have the same DNF.

Method 1 (Truth Table Approach):

1. Two propositions are equivalent if their truth tables are identical.
2. The Disjunctive Normal Form (DNF) of a proposition is constructed by taking an OR of all minterms corresponding to truth assignments where the proposition evaluates to True.
3. If two propositions are equivalent, they share the exact same rows marked True in their truth tables.
4. Therefore, their DNF expressions (built from these rows) will be identical.

Method 2 (Algebraic/Uniqueness of DNF):

1. Every proposition has a unique canonical DNF (important theorem) determined by its truth table.
2. If two propositions U and V are equivalent ( $U \equiv V$ ), then for every assignment, U and V take the same truth value.
3. Hence, the canonical DNF constructed from U will be the same as the canonical DNF constructed from V.

4. Therefore, equivalent propositions always have the same DNF.

Thus, the statement is proven.

## Improved Full-Marks Solution for Q2

Q2. Prove: If two propositions are equivalent then they have the same DNF.

Given: Propositions P and Q satisfy  $P \equiv Q$  (same truth tables) (i.e. they have the same truth value under every assignment).

To prove: The canonical Disjunctive Normal Forms (DNFs) of P and Q are identical.

**[Key step]**

### Method A: Truth Table / Canonical DNF Construction

1. The canonical DNF of a proposition is obtained by listing every row of its truth table where the formula evaluates to True. For each such row, form an AND-term that matches the row (variable unnegated if True, negated if False). The canonical DNF is the OR of all such terms.
2. Since  $P \equiv Q$  (same truth tables), their truth tables are identical. In particular, the set of rows where P is True equals the set of rows where Q is True.
3. Therefore, the same AND-terms are generated when forming  $DNF(P)$  and  $DNF(Q)$ .
4. Hence  $DNF(P) = DNF(Q)$  [Therefore proven]. ■ [Key step]

### Method B: Uniqueness of Canonical DNF

1. Every Boolean function has a unique canonical DNF (important theorem) (sum of minterms) determined by its truth table.
2. If  $P \equiv Q$  (same truth tables), then P and Q represent the same Boolean function (same truth table).
3. By uniqueness,  $DNF(P) = DNF(Q)$  [Therefore proven].
4. Therefore equivalent propositions always have the same DNF. ■ [Key step]

### Illustrative Example

Let  $A = \neg p \vee q$  and  $B = p \Rightarrow q$ .

Their truth tables are identical:

| p | q | A | B |
|---|---|---|---|
| T | T | T | T |
| T | F | F | F |
| F | T | T | T |
| F | F | T | T |

Rows where the formulas are True:  $(p=T, q=T)$ ,  $(p=F, q=T)$ ,  $(p=F, q=F)$ .

Constructing canonical DNF from these rows yields the same expression for both A and B.

Thus,  $DNF(A) = DNF(B)$ , confirming the theorem.

## Q1.2 Rubric

**Q1.2 - 1) (6 marks)**

Each true or false question (1.5 marks)

- 1.5 marks – Correct final answer and justification.
- 0.5 marks – Correct final answer without justification
- 0.00 marks – No valid attempt.

**Q1.2 - 2) Rubric (4 marks)**

- 4 marks – Correct final answer and justification.
- 2 - 3 marks – correct final answer and some mistakes in justification.
- 1 marks – correct final answer but no justification.
- 0.00 marks – No valid attempt.

**Solutions****Q1.2 - 1) Solution**

(a) Let  $x$  be a variable. Then, " $x + 5 > 10$ " is a proposition.

**False.**

A proposition is a declarative statement that has a definite truth value, meaning it is either true or false, but not both. The statement " $x + 5 > 10$ " contains a variable,  $x$ . The truth value of this statement depends on the value assigned to  $x$ . For example, if  $x=6$ , the statement is true ( $11 > 10$ ), but if  $x=2$ , the statement is false ( $7 > 10$ ). Since its truth value is not fixed, it is a predicate (or an open sentence), not a proposition.

(b)  $(p \wedge q) \vee (\neg p \wedge \neg q)$  is a tautology, where  $p, q$  are propositional variables.

**False.**

A tautology is a statement that is true for all possible truth values of its propositional variables. We can determine if this statement is a tautology by constructing a truth table.

| $p$ | $q$ | $p \wedge q$ | $\neg p$ | $\neg q$ | $\neg p \wedge \neg q$ | $(p \wedge q) \vee (\neg p \wedge \neg q)$ |
|-----|-----|--------------|----------|----------|------------------------|--------------------------------------------|
| T   | T   | T            | F        | F        | F                      | T                                          |
| T   | F   | F            | F        | T        | F                      | F                                          |
| F   | T   | F            | T        | F        | F                      | F                                          |
| F   | F   | F            | T        | T        | T                      | T                                          |

Since the final column contains both 'T' and 'F' values, the statement is not always true. Therefore, it is not a tautology; it is a contingency. This expression is logically equivalent to the biconditional  $p \Leftrightarrow q$ .

(c)  $p \Leftrightarrow q \equiv \neg(p \oplus q)$ , where  $p, q$  are propositional variables.

**True.**

Two compound propositions are logically equivalent if they have the same truth table. Let's construct the truth tables for both  $p \Leftrightarrow q$  (biconditional) and  $\neg(p \oplus q)$  (negation of exclusive OR).

Truth table for  $p \Leftrightarrow q$ :

| $p$ | $q$ | $p \Leftrightarrow q$ |
|-----|-----|-----------------------|
| T   | T   | T                     |
| T   | F   | F                     |
| F   | T   | F                     |
| F   | F   | T                     |

Truth table for  $\neg(p \oplus q)$ :

| $p$ | $q$ | $p \oplus q$ | $\neg(p \oplus q)$ |
|-----|-----|--------------|--------------------|
| T   | T   | F            | T                  |
| T   | F   | T            | F                  |
| F   | T   | T            | F                  |
| F   | F   | F            | T                  |

The final truth value columns for both expressions are identical (T, F, F, T). Thus, they are logically equivalent.

(d) Let  $x$  be a variable and  $P(x)$  be the predicate  $x^2 < x + 3$ . Suppose the domain of  $P(x)$  is all natural numbers. Then,  $\exists x P(x)$  is true.

**True.**

The statement  $\exists x P(x)$  means "there exists at least one natural number  $x$  for which the predicate  $P(x)$  is true." The domain is the set of natural numbers ( $N=\{1,2,3,\dots\}$ ). We only need to find one value of  $x$  that satisfies the inequality  $x^2 < x+3$ .

Let's test the first few natural numbers:

- For  $x = 1$ :  $1^2 < 1+3 \Rightarrow 1 < 4$ . This is True.
- For  $x = 2$ :  $2^2 < 2+3 \Rightarrow 4 < 5$ . This is True.

Since we have found at least one natural number (e.g.,  $x=1$ ) for which  $P(x)$  is true, the existential quantification  $\exists x P(x)$  is true.

### Q1.2 - 2) Solution

The statement  $\forall x \exists y P(x,y)$  translates to: "For every natural number  $x$ , there exists a natural number  $y$  such that  $P(x,y)$  is true."

For the entire statement to be true, for any  $x$  we choose, we must be able to find at least one  $y$  that makes the predicate  $P(x,y)$  true. To prove the statement is false, we only need to find a single counterexample for  $x$ —that is, one value of  $x$  for which there is no possible  $y$  that makes  $P(x,y)$  true.

Let's use  $x = 1$  as a counterexample.

For  $x=1$ , the statement becomes  $\exists y P(1,y)$ . Let's evaluate the predicate  $P(1,y)$ :  
 $P(1,y) \equiv (1^2 + 3 \leq 13) \wedge (y^2 - 1 \geq y)$   $P(1,y) \equiv (1 + 3 \leq 1) \wedge (y^2 - 1 \geq y)$   $P(1,y) \equiv (4 \leq 1) \wedge (y^2 - 1 \geq y)$

The first part of the conjunction,  $4 \leq 1$ , is False. In a logical AND ( $\wedge$ ) operation, if one part is false, the entire statement is false, regardless of the truth value of the other part.

Therefore, for  $x=1$ , the predicate  $P(1,y)$  is false for all possible values of  $y$ . This means that  $\exists y P(1,y)$  is false.

Since we have found a value for  $x$  (namely  $x=1$ ) for which the statement " $\exists y P(x,y)$ " does not hold, the universally quantified statement  $\forall x \exists y P(x,y)$  is **False**.

### Q1.3 Rubric

Q1.3.1:

1. If solved using a Truth Table (T.T):

T.T (4 marks,  $\frac{1}{2}$  mark each)

CNF (2 marks)

DNF (2 marks)

2. If solved using Propositional Logic (P.L):

CNF (4 marks)

DNF (4 marks)

3. Mixed approach:

CNF solved using P.L  $\rightarrow$  4 marks

DNF solved using T.T  $\rightarrow$  (T.T: 2 marks (0.25 marks each), DNF: 2 marks)

(In Truth Table marks are given only for  $p$ ,  $q$ ,  $r$ ,  $\sim q$ ,  $\sim r$ ,  $(p \vee \sim q)$ ,  $\sim(p \vee \sim q)$ ,  $\sim(p \vee \sim q) \Rightarrow \sim r$ )

(1 partial mark is given for both CNF and DNF in case the written CNF and DNF satisfy the Conditions of CNF and DNF).

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Q2:

If the logic of  $q \rightarrow p$  is explained using words or a truth table (T.T.),  
or if it is explained that when  $q = \text{true}$ ,  $p$  can be both true or false,  
or if an explanation of  $q$  being an conjecture is given  
then 1 mark is awarded (for any of these explanations).

If it is stated that the truth value of  $q \rightarrow p$  cannot be inferred because  $p$  is unknown, 1 mark is awarded.

Additional note:

1. If  $p = \text{false}$  is provided as a solution, it is also accepted provided an explanation is given.

i.e.

$q = \text{true}$ ,  $p = \text{false}$

$q \rightarrow p = \text{false}$

With Proper Explanation.

2. Partial Marks / No marks are given in case the explanation is wrong or vague.

**Question 1.4.** 1. (4 marks). Let  $x, y$  be variables and  $P(x, y), Q(x, y)$  be propositions. Simplify the following using rules of predicate and propositional logic

$$\neg \forall x \exists y (P(x, y) \wedge Q(x, y)).$$

**Solution: Step 1: Apply De Morgan's Law for the Universal Quantifier**

We start with the expression:

$$\neg \forall x \exists y (P(x, y) \wedge Q(x, y))$$

Using the rule:

$$\neg \forall x R(x) \equiv \exists x \neg R(x)$$

where  $R(x) = \exists y (P(x, y) \wedge Q(x, y))$ , we obtain:

$$\neg \forall x \exists y (P(x, y) \wedge Q(x, y)) \equiv \exists x \neg (\exists y (P(x, y) \wedge Q(x, y)))$$

**Step 2: Apply De Morgan's Law for the Existential Quantifier**

Now, simplify the inner expression:

$$\neg (\exists y (P(x, y) \wedge Q(x, y)))$$

Using the rule:

$$\neg \exists y S(y) \equiv \forall y \neg S(y)$$

where  $S(y) = P(x, y) \wedge Q(x, y)$ , we get:

$$\neg (\exists y (P(x, y) \wedge Q(x, y))) \equiv \forall y \neg (P(x, y) \wedge Q(x, y))$$

**Step 3: Apply De Morgan's Law for the Conjunction**

Simplify the expression inside the universal quantifier:

$$\neg (P(x, y) \wedge Q(x, y))$$

Using the rule:

$$\neg (A \wedge B) \equiv \neg A \vee \neg B$$

we obtain:

$$\neg (P(x, y) \wedge Q(x, y)) \equiv \neg P(x, y) \vee \neg Q(x, y)$$

Thus:

$$\forall y \neg (P(x, y) \wedge Q(x, y)) \equiv \forall y (\neg P(x, y) \vee \neg Q(x, y))$$

**Step 4: Combine All Steps**

Substituting back, we have:

$$\neg \forall x \exists y (P(x, y) \wedge Q(x, y)) \equiv \exists x \forall y (\neg P(x, y) \vee \neg Q(x, y))$$

**Final Simplified Form:**

$$\boxed{\exists x \forall y (\neg P(x, y) \vee \neg Q(x, y))}$$

**Marking Scheme:** +1 for correct expression, +2 mention of necessary rules, +1 step by step Calculation.

2. (4 marks). Suppose we define a new quantifier  $\varphi$  as follows:

**Definition 1.** Let  $x$  be a variable and  $P(x)$  be a predicate. Then,  $\varphi xP(x)$  denote the statement “there exist at least two values  $a, b$  in the domain of  $P(x)$  such that  $P(a), P(b)$  are true”.

(a) Is  $\varphi xP(x)$  a proposition? Justify your answer.

(b) Is  $\varphi xP(x) \equiv \exists xP(x)$  always? **Solution:**

**Step 1: Analyze Part (a)**

The quantifier  $\varphi$  is defined as:

$\varphi xP(x) :=$  “There exist at least two values  $a, b$  in the domain such that  $P(a)$  and  $P(b)$  are true.”

A **proposition** is a statement that is either true or false. Since  $\varphi xP(x)$  asserts a specific condition about the predicate  $P(x)$  over its domain, its truth value can be determined once the domain and  $P(x)$  are specified. This is analogous to standard quantifiers like  $\forall$  and  $\exists$ , which also yield propositions. Therefore,  $\varphi xP(x)$  is indeed a proposition.

Yes,  $\varphi xP(x)$  is a proposition.

**Step 2: Analyze Part (b)**

We are asked whether  $\varphi xP(x) \equiv \exists xP(x)$  always holds.

- $\exists xP(x)$  means “There exists at least one value in the domain for which  $P(x)$  is true.”
- $\varphi xP(x)$  means “There exist at least two distinct values in the domain for which  $P(x)$  is true.”

These are not equivalent. If only one element satisfies  $P(x)$ , then  $\exists xP(x)$  is true, but  $\varphi xP(x)$  is false.

**Counterexample:**

Let the domain be the set of integers  $\mathbb{Z}$ , and define the predicate:

$$P(x) := x = 0$$

- $\exists xP(x)$  is true because  $P(0)$  holds.
- $\varphi xP(x)$  is false because only  $x = 0$  satisfies  $P(x)$ ; there is no second distinct value  $b$  such that  $P(b)$  is true.

Thus,  $\varphi xP(x) \not\equiv \exists xP(x)$ .

No,  $\varphi xP(x) \not\equiv \exists xP(x)$  always.

**Final Answers:**

(a) Yes,  $\varphi xP(x)$  is a proposition.

(b) No,  $\varphi xP(x) \not\equiv \exists xP(x)$  always.

**Marking Scheme:** (a) +1 for correct answer and +1 for proper justification. (b) +1 for correct answer and +1 for correct counter example.

3. (2 marks). Mention your favourite topic in mathematical logic.

**Marking Scheme:** Binary Marking.